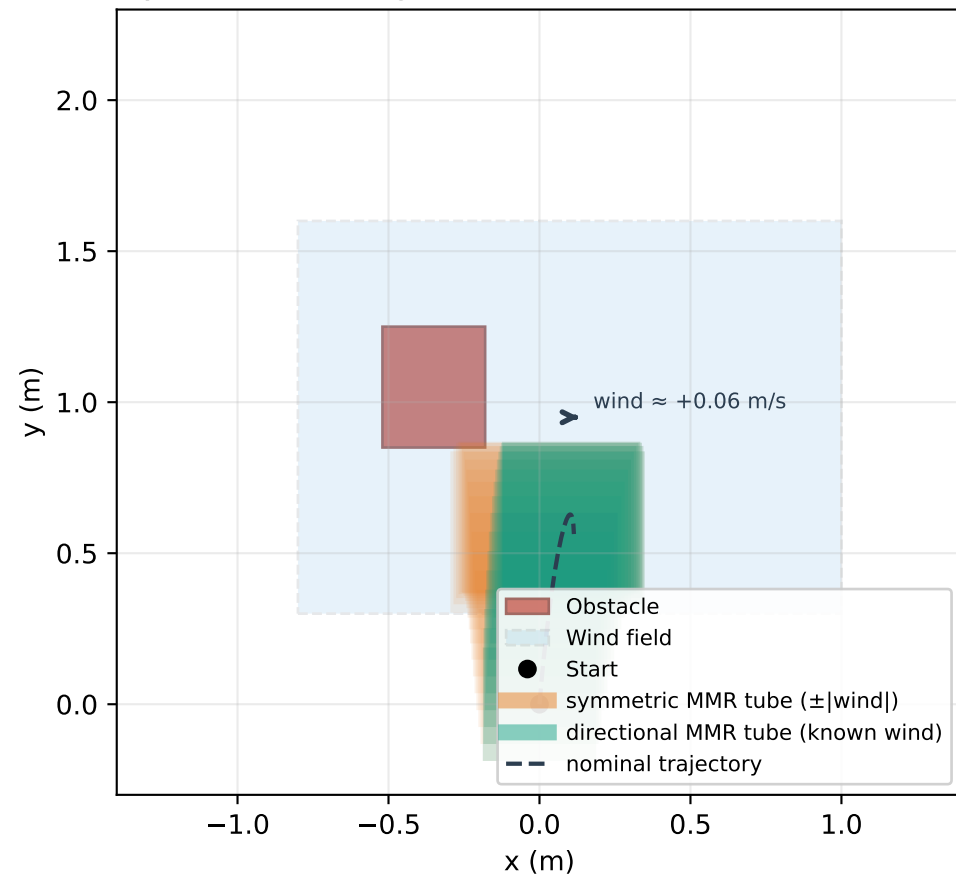


Scenario: wind — Known wind in +x; obstacle to the left. Symmetric tube model still extends $\pm|\text{wind}|$ left, projecting into the obstacle (false alarm). Directional tube model knows the wind shifts the tube right, away from the obstacle (true safe).

Initial $k = (-0.12, 0.85)$
symmetric (orange) vs directional (teal) MMR tubes



Final k from each method
MMR tube envelope + nominal trajectory

